

Reference : David Tong, Chapter 2 & Peskin, Chapter 2

A. Zee, Chapter 1 & Nagaosa, Chapter 2 & Atland, Chapter 4

Content : 1. Canonical Quantization

- Scalar field
- Complex scalar field
- Invariant and conservation

2. Causality and Dynamics

- Heisenberg picture
- Causality and propagator
- Recovery of the known

3. Path Integral Approach

- Quantum Mechanics
- Mathematical foundation
- Quantum field theory

4. Application

- Simplified casimir effect

Part 1.

Quantum field theory is an operator valued function.

$$\begin{cases} [\phi_a(\vec{x}), \phi_b(\vec{y})] = [\pi^a(\vec{x}), \pi^b(\vec{y})] = 0 \\ [\phi_a(\vec{x}), \pi^b(\vec{y})] = i \delta(\vec{x} - \vec{y}) \delta_a^b \end{cases} \quad \begin{array}{l} \text{Treating time/space on} \\ \text{different footing} \end{array}$$

Free field: each degree of freedom evolves independently

(easy to obtain the spectrum; Lagrangians usually quadratic)

KG field: $\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2$

$$\partial_\mu \partial^\mu \phi + m^2 \phi = 0 \xrightarrow[\text{Transformation}]{\text{Fourier}} \left[\frac{\partial^2}{\partial t^2} + \omega(\vec{p})^2 \right] \phi(\vec{p}, t) = 0 \quad \begin{array}{l} \text{free field} \\ \omega^2 = p^2 + m^2 \end{array}$$

General Solution:

$$\begin{cases} \phi(\vec{p}, t) = \phi_1(\vec{p}) e^{-i\omega_p t} + \phi_2(\vec{p}) e^{i\omega_p t} \\ \phi(\vec{x}, t) = \int \frac{d^3 p}{(2\pi)^3} e^{i\vec{p} \cdot \vec{x}} \phi(\vec{p}, t) \end{cases}$$

$$\text{Hermitian} \rightarrow \phi(\vec{p}, t) = \phi^\dagger(-\vec{p}, t) \rightarrow \phi_2(\vec{p}) = \phi_1^\dagger(-\vec{p})$$

$$\Rightarrow \begin{cases} \phi(\vec{p}, t) = \phi_1(\vec{p}) e^{-i\omega_p t} + \phi_1^\dagger(-\vec{p}) e^{i\omega_p t} \\ \pi(\vec{p}, t) = -i\omega_p \phi_1(\vec{p}) e^{-i\omega_p t} + i\omega_p \phi_1^\dagger(-\vec{p}) e^{i\omega_p t} \end{cases}$$

Define $a_p = \sqrt{2\omega_p} \phi_1(\vec{p})$, the commutation relations become:

$$\begin{cases} [a_p, a_q] = [a_p^\dagger, a_q^\dagger] = 0 \\ [a_p, a_q^\dagger] = (2\pi)^3 \delta(\vec{p} - \vec{q}) \end{cases} \rightarrow \text{Recall simple harmonic oscillator}$$

Ultimately, we have:

$$\begin{cases} \phi(\vec{x}) = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} [a_p e^{i\vec{p} \cdot \vec{x}} + a_p^\dagger e^{-i\vec{p} \cdot \vec{x}}] \\ \pi(\vec{x}) = \int \frac{d^3 p}{(2\pi)^3} (-i) \sqrt{\frac{\omega_p}{2}} [a_p e^{i\vec{p} \cdot \vec{x}} - a_p^\dagger e^{-i\vec{p} \cdot \vec{x}}] \end{cases}$$

$$\text{or} \begin{cases} \phi(\vec{x}, t) = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} [a_p e^{i\vec{p} \cdot \vec{x}} + a_p^\dagger e^{-i\vec{p} \cdot \vec{x}}] \\ \pi(\vec{x}, t) = \int \frac{d^3 p}{(2\pi)^3} (-i) \sqrt{\frac{\omega_p}{2}} [a_p e^{i\vec{p} \cdot \vec{x}} - a_p^\dagger e^{-i\vec{p} \cdot \vec{x}}] \end{cases}$$

← Discuss Later

Calculate the Hamiltonian:

$$H = \int \frac{d^3p}{(2\pi)^3} \omega_p [a_p^\dagger a_p + \frac{1}{2} (2\pi)^3 \delta(0)]$$

Two types of infinities: infra-red divergence / ultra-violet divergence

$$(2\pi)^3 \delta(0) = \lim_{L \rightarrow \infty} \int_{-L/2}^{L/2} d^3x e^{i\vec{x} \cdot \vec{p}} \Big|_{\vec{p} \rightarrow 0} = V$$

ground state energy
cut-off at high momentum

Normal Ordering: $:H: = \int \frac{d^3p}{(2\pi)^3} \omega_p a_p^\dagger a_p$
spectrum

cosmological constant?

Complex scalar field: $\mathcal{L} = \partial_\mu \phi^* \partial^\mu \phi - \mu^2 \phi^* \phi$

EoM: $\begin{cases} \partial_\mu \partial^\mu \phi + \mu^2 \phi = 0 \\ \partial_\mu \partial^\mu \phi^* + \mu^2 \phi^* = 0 \end{cases}$ No Hermitian requirement, we get:

$$\begin{cases} \phi(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} (b_p e^{i\vec{p} \cdot \vec{x}} + c_p^\dagger e^{-i\vec{p} \cdot \vec{x}}) \\ \pi^\dagger(x) = \dot{\phi} = \int \frac{d^3p}{(2\pi)^3} (i) \sqrt{\frac{\omega_p}{2}} (b_p e^{i\vec{p} \cdot \vec{x}} - c_p^\dagger e^{-i\vec{p} \cdot \vec{x}}) \end{cases}$$

Commutation Relation gives: (Non-trivial part)

$$[b_p, b_q^\dagger] = [c_p, c_q^\dagger] = (2\pi)^3 \delta(\vec{p} - \vec{q})$$

Calculate the Hamiltonian:

$$\begin{aligned} H &= \int d^3x \pi \pi^\dagger + \pi^\dagger \pi - \pi \pi^\dagger + |\nabla \phi|^2 + \mu^2 \phi^\dagger \phi \\ &= \int d^3x \frac{d^3p d^3q}{(2\pi)^6} \left(-\frac{\sqrt{\omega_p \omega_q}}{2} \right) (b_p - c_p^\dagger) (b_q^\dagger - c_q) e^{i(\vec{p} + \vec{q}) \cdot \vec{x}} + \frac{\vec{p} \cdot \vec{q} + \mu^2}{2\sqrt{\omega_p \omega_q}} (b_p + c_p^\dagger) (b_q^\dagger + c_q) e^{i(\vec{p} + \vec{q}) \cdot \vec{x}} \\ &= \int \frac{d^3p}{(2\pi)^3} \frac{\omega_p}{2} [(b_p + c_{-p}^\dagger) (b_{-p}^\dagger + c_p) - (b_p - c_{-p}^\dagger) (b_{-p}^\dagger - c_p)] \\ &= \int \frac{d^3p}{(2\pi)^3} \omega_p (b_p^\dagger b_p + c_p^\dagger c_p) \end{aligned}$$

Particles and invariant

Def: the vacuum: $a_{\vec{p}} |0\rangle = 0, \forall \vec{p}, \langle 0|0\rangle = 1$

For scalar field, we have: $[H, a_p^\dagger] = \omega_p a_p^\dagger, [H, a_p] = -\omega_p a_p$

The energy eigenstates $|\vec{p}\rangle = a_p^\dagger |0\rangle, H|\vec{p}\rangle = \omega_p |\vec{p}\rangle$

→ Create multi-particles states in Fock space.

BUT:

(a) the eigenstates is not particles:

$$\langle \vec{p} | \vec{p} \rangle = \langle 0 | a_p a_p^\dagger | 0 \rangle = (2\pi)^3 \delta(0), \quad \langle 0 | \phi(x) \phi^\dagger(x) | 0 \rangle = \langle x | x \rangle = \delta(0)$$

They are operator valued distributions! We should create wave packets!

(b) Does the distributions Lorentz invariant?

We prefer under Lorentz transformation, $|p\rangle \rightarrow |p'\rangle = U(\Lambda)|p\rangle$

where $U(\Lambda)$ is unitary $\rightarrow$ relativistic normalization

Construct invariant!

$$\textcircled{1} I = \int \frac{d^3p}{(2\pi)^3} |p\rangle \langle p| \quad \textcircled{2} \int d^3p \delta(p_0^2 - p^2 - m^2) |p_0\rangle = \int \frac{d^3p}{2p_0} |p_0 = E_p\rangle = \int \frac{d^3p}{2E_p}$$

$\Rightarrow$ relativistically normalized state is $\sqrt{2E_p} |p\rangle$, the creation operator $a_p^\dagger = \sqrt{2E_p} a_p^\dagger$

Conservation law: Classical / Quantum?

Translation symmetry of scalar field gives:

$$\begin{aligned} p^i &= -\int d^3x \phi \partial^i \phi \\ &= -\int d^3x \frac{d^3p}{(2\pi)^3} \frac{d^3q}{(2\pi)^3} (-i) \sqrt{\omega_q/\omega_p} \frac{1}{2} (a_q e^{i\vec{q}\cdot\vec{x}} - a_q^\dagger e^{-i\vec{q}\cdot\vec{x}}) (i p^i a_p e^{i\vec{p}\cdot\vec{x}} - i p^i a_p^\dagger e^{-i\vec{p}\cdot\vec{x}}) \\ &= -\int \frac{d^3x}{2(2\pi)^3} \frac{d^3p}{2} \frac{d^3q}{2} p^i \sqrt{\omega_q/\omega_p} [a_q a_p e^{i(\vec{q}+\vec{p})\cdot\vec{x}} + a_q^\dagger a_p^\dagger e^{-i(\vec{q}+\vec{p})\cdot\vec{x}} - a_q a_p^\dagger e^{i(\vec{q}-\vec{p})\cdot\vec{x}} - a_q^\dagger a_p e^{-i(\vec{q}-\vec{p})\cdot\vec{x}}] \\ &= -\int \frac{1}{2} \frac{d^3p}{(2\pi)^3} p^i [\underbrace{a_p a_{-p} + a_{-p}^\dagger a_p^\dagger}_{\text{cancel}} - \underbrace{a_p a_p^\dagger - a_p^\dagger a_p}_{\text{normal ordering}}] \\ &= \int \frac{d^3p}{(2\pi)^3} p^i a_p^\dagger a_p \quad [p^i, H] = 0! \end{aligned}$$

$\Rightarrow |p\rangle$ is the momentum eigenstates

Rotation symmetry: for scalar field,

$$\begin{aligned} Q^{ij} &= \int d^3x (x^i T^{0j} - x^j T^{0i}) = \int d^3x (x^i \partial^0 \phi \partial^j \phi - x^j \partial^0 \phi \partial^i \phi) \\ &= \int d^3x \frac{d^3p}{2(2\pi)^3} \frac{d^3q}{2} (x^i p^j - x^j p^i) \sqrt{\frac{\omega_q}{\omega_p}} e^{i\vec{q}+\vec{q}\cdot\vec{x}} (a_p + a_{-p}^\dagger)(a_q - a_{-q}^\dagger) \\ &= \int d^3x \frac{d^3p}{2(2\pi)^3} \frac{d^3q}{2} (x^i p^j - x^j p^i) \sqrt{\frac{\omega_q}{\omega_p}} e^{i\vec{q}+\vec{q}\cdot\vec{x}} (a_p a_q + a_{-p}^\dagger a_{-q}^\dagger - a_{-p}^\dagger a_q - a_p a_{-q}^\dagger) \end{aligned}$$

$$Q^{ij} |p=0\rangle = ?$$

Number operator $N = \int \frac{d^3p}{(2\pi)^3} a_p^\dagger a_p$ for $[N, H] = 0 \xrightarrow{\text{dynamic}}$ "conserved!"

For complex scalar field, we have classical conserved charge

$$\begin{cases} Q = i \int d^3x (\pi \psi - \psi^* \pi^*) = \int \frac{d^3p}{(2\pi)^3} (c_p^\dagger c_p - b_p^\dagger b_p) \\ [Q, H] = [\int \frac{d^3p}{(2\pi)^3} (c_p^\dagger c_p - b_p^\dagger b_p), \int \frac{d^3q}{(2\pi)^3} \omega_q (b_q c_q + b_q^\dagger c_q^\dagger)] = 0 \end{cases}$$

Notice: Q is still conserved in interacting theory, N_c/N_b is conserved in free field.

Heisenberg picture:

To present Lorentz invariant, we introduce Heisenberg picture:

$$\frac{dO_H}{dt} = i[H, O_H] \rightarrow \text{still apply in QFT}$$

It's straight forward to calculate:

$$\begin{cases} \dot{\phi} = i [\pi, \phi] = \pi(x) \\ \dot{\pi} = \nabla^2 \phi - m^2 \phi \end{cases} \Rightarrow \partial_\mu \partial^\mu \phi + m^2 \phi = 0 \quad \text{KG equation!}$$

As usual, we have $a_p^\dagger, a_p$ evolve

$$\begin{cases} e^{iHt} a_p^\dagger e^{-iHt} = e^{-iE_p t} a_p^\dagger \\ e^{iHt} a_p e^{-iHt} = e^{iE_p t} a_p \end{cases} \Rightarrow \phi(x, 0) = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} (a_p e^{ip \cdot x} + a_p^\dagger e^{-ip \cdot x})$$

Causality: the measure of one place will influence another one?

Define $\Delta(x-y) = [\phi(x), \phi(y)]$

$$\begin{aligned} &= \int \frac{d^3 p d^3 q}{(2\pi)^6} \frac{1}{\sqrt{4E_p E_q}} [a_p e^{ip \cdot x} + a_p^\dagger e^{-ip \cdot x}, (a_q e^{iq \cdot y} + a_q^\dagger e^{-iq \cdot y})] \\ &= \int \frac{d^3 p d^3 q}{(2\pi)^6} \frac{1}{\sqrt{4E_p E_q}} ([a_p, a_q^\dagger] e^{ip \cdot x - iq \cdot y} + [a_p^\dagger, a_q] e^{-ip \cdot x + iq \cdot y}) \\ &= \int \frac{d^3 p}{(2\pi)^3} \frac{1}{2E_p} [e^{-ip \cdot (x-y)} - e^{ip \cdot (x-y)}] \end{aligned}$$

Check:

(a) Lorentz invariant!

(b) Doesn't vanish for time-like separation!

(c) vanish for space-like separation!

↓
equal time separation

Propagator: the road to path integral!

$$\langle 0 | \phi(x) \phi(y) | 0 \rangle = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{2E_p} e^{-ip \cdot (x-y)} \equiv D(x-y) \longrightarrow \text{propagator!}$$

for $(x-y)^2 < 0$ (time-like!)

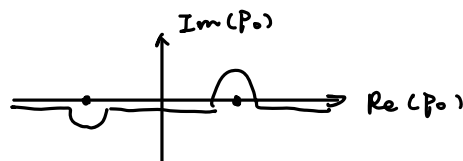
we can know: $D(x-y) \sim e^{-m|x-y|}$

Define Feynman propagator:

$$\Delta_F(x-y) = \langle 0 | T \phi(x) \phi(y) | 0 \rangle = \begin{cases} D(x-y) & x^0 > y^0 \\ D(y-x) & x^0 < y^0 \end{cases}$$

$$= \int \frac{d^4 p}{(2\pi)^4} \frac{i}{p^2 - m^2} e^{-ip \cdot (x-y)} \longrightarrow \text{to show this, use "ie prescription"!$$

the contour is



As Green's function: If stay away from singularities,

$$(\partial_\mu^2 - \nabla^2 + m^2) \Delta_F(x-y) = -i \delta^4(x-y)$$

Retarded/advanced Green's function:

$$\Delta_R(x-y) = \begin{cases} D(x-y) - D(y-x) & x^0 > y^0 \\ 0 & x^0 < y^0 \end{cases}$$

Recovery of the known:

we decompose the field: $\psi(\vec{x}, t) = e^{-imt} \tilde{\psi}(\vec{x}, t)$

then KG equation becomes: $e^{-imt} [\ddot{\tilde{\psi}} - 2im\dot{\tilde{\psi}} - \nabla^2 \tilde{\psi}] = 0$

Non-relativistic limit is $|\vec{p}| \ll m$, $\ddot{\tilde{\psi}}$ omit!

$$i \frac{\partial \tilde{\psi}}{\partial t} = -\frac{1}{2m} \nabla^2 \tilde{\psi} \quad \text{No probability problem!}$$

For a complex scalar field,

$$\mathcal{L} = \partial_\mu \phi^* \partial^\mu \phi - m^2 \phi^* \phi \quad \phi = e^{-imt} \tilde{\phi} \quad \phi^* = e^{imt} \tilde{\phi}^*$$

$$\Rightarrow \mathcal{L} = i \psi^\dagger \dot{\psi} - \frac{1}{2m} \nabla \psi^* \cdot \nabla \psi = \frac{i}{2} (\phi^* \dot{\phi} - \dot{\phi}^* \phi) - \frac{1}{2m} \nabla \psi^* \cdot \nabla \psi$$

$$\begin{aligned} \partial_\mu \phi^* \partial^\mu \phi &= (im \phi^* + \dot{\phi}^*) (-im \phi + \dot{\phi}) \\ &\approx m^2 \phi \phi^* + im(\phi^* \dot{\phi} - \dot{\phi}^* \phi) \end{aligned}$$

$$j^\mu = (-\psi^* \dot{\psi}, \frac{i}{2m} (\psi^* \nabla \psi - \nabla \psi^* \psi))$$

Then we quantise this field as before, we find

$$\psi(\vec{x}) = \int \frac{d^3p}{(2\pi)^3} a_p e^{i\vec{p}\cdot\vec{x}}, \quad [a_p, a_q^\dagger] = (2\pi)^3 \delta(\vec{p}-\vec{q}), \quad \psi^\dagger(\vec{x}) = \int \frac{d^3p}{(2\pi)^3} a_p^\dagger e^{-i\vec{p}\cdot\vec{x}}$$

$$H = \frac{1}{2m} \nabla \psi^* \cdot \nabla \psi = \int \frac{d^3p}{(2\pi)^3} \frac{p^2}{2m} a_p^\dagger a_p$$

What should be noticed here:

$$[a_p, \psi^\dagger(\vec{x})] = e^{-i\vec{p}\cdot\vec{x}}$$

1) No anti-particle $\rightarrow$ requirement of relativity!

2) $\int d^3x \psi^* \psi$ is conserved $\rightarrow$ particle number!

3) No non-relativistic limit of a real scalar field!

$$\text{Construct } \begin{cases} X = \int d^3x \psi^\dagger(x) \psi(x) \\ P = \int d^3p / (6\pi)^3 \vec{p} a_p^\dagger a_p \end{cases} \quad \text{assume } |\varphi\rangle = \int d^3x \varphi(\vec{x}) |\vec{x}\rangle$$

$$P^i |\varphi\rangle = \int \frac{d^3x d^3p}{(2\pi)^3} \varphi(x) p^i a_p^\dagger a_p \varphi(\vec{x}) \psi^\dagger(x) |0\rangle$$

$$= \int \frac{d^3x d^3p}{(2\pi)^3} \varphi(x) p^i a_p^\dagger e^{-i\vec{p}\cdot\vec{x}} |0\rangle$$

$$= \int \frac{d^3x d^3p}{(2\pi)^3} \varphi(x) (i \frac{\partial}{\partial x^i} e^{i\vec{p}\cdot\vec{x}}) a_p^\dagger |0\rangle$$

$$= \int \frac{d^3x d^3p}{(2\pi)^3} e^{i\vec{p}\cdot\vec{x}} (-i \frac{\partial}{\partial x^i} \varphi(x)) a_p^\dagger |0\rangle$$

$$= \int \frac{d^3x}{(2\pi)^3} (-i \frac{\partial \varphi}{\partial x^i}) |x\rangle$$

$$\Rightarrow [X^i, P^j] = i \delta^{ij}$$

Dynamic of $\varphi(x, t)$:

$$\dot{\psi}^\dagger = i [H, \psi^\dagger] = i \int \frac{d^3p}{(2\pi)^3} \frac{p^2}{2m} [a_p^\dagger a_p, \psi^\dagger] = i \int \frac{d^3p}{(2\pi)^3} \frac{p^2}{2m} a_p^\dagger e^{-i\vec{p}\cdot\vec{x}}$$

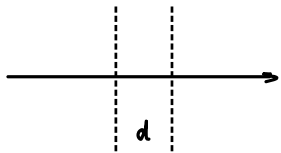
$$\varphi(x, t) \psi_x^\dagger = \varphi(x) \psi_x^\dagger$$

$$\frac{\partial \varphi}{\partial t} \psi^\dagger = \varphi \dot{\psi}^\dagger = i \int \frac{d^3p}{(2\pi)^3} \varphi\left(-\frac{1}{2m}\right) v^2 e^{-i\vec{p}\cdot\vec{x}} a_p^\dagger$$

$$= \frac{i}{2m} \nabla^2 \varphi \psi^\dagger$$

Then $i \frac{\partial \varphi}{\partial t} = -\frac{1}{2m} \nabla^2 \varphi \longrightarrow$ Schrödinger equation!

Casimir effect: vacuum fluctuation $\longleftarrow$ simplification $\left\{ \begin{array}{l} 1+1d \\ \text{scalar field} \\ \text{massless} \end{array} \right.$



Q: What does it mean " $\phi(x)=0$ "?

The plates quantize the momentum: $p_x = \frac{n\pi}{d}$

$$\text{Then } E_{1+1}(d) = \frac{\pi}{2d} \sum_{n=1}^{\infty} n e^{-an\pi/d} \xrightarrow{\text{UV cut-off}} \\ = \frac{d}{2\pi a^2} - \frac{\pi}{24d} + \mathcal{O}(a^2)$$

$$\Rightarrow E_{\text{total}} = \underbrace{E_{1+1}(d) + E_{1+1}(L-d)}_{\text{infra-red}} = \frac{L}{2\pi a^2} - \frac{\pi}{24} \left(\frac{1}{d} + \frac{1}{L-d} \right) + \mathcal{O}(a^2) \longrightarrow \frac{\partial E}{\partial d} = \frac{\pi}{24d^2} + \dots$$

$$\text{In } 3+1 \text{ dimensions, } \frac{1}{A} \frac{\partial E}{\partial d} = \frac{\pi^2}{480d^4}$$